

Eren Ege Çelik

İzmir, Turkey | erenege3500@gmail.com | erenege.com | github.com/ErenEgeCelik

EDUCATION

Middle East Technical University (ODTÜ)

2024 – 2028 (expected)

BSc Physics · Ankara, Turkey · Penultimate year 2026–27, on track to graduate June 2028

İzmir Institute of Technology (İYTE)

Physics (transferred to ODTÜ) · GPA 3.6 / 4.0 · İzmir, Turkey

HONORS & SELECTION

- İzmir Mathematics Olympiad Team — selected to the regional team, grades 8–9.
- Physics Olympiad Team, İzmir Atatürk High School (among Turkey's top 5 high schools) — final year.
- Admitted to İzmir Atatürk High School through nationally competitive examination (top ~0.2%).

EXPERIENCE

Independent Quantitative Researcher & Trader — Prediction Markets

2026 – present

Self-directed; live trading on Polymarket · Remote

- Grew a single \$30 deposit to roughly \$1,200 in net P&L over three months of live trading, with no additional capital injection.
- Reverse-engineered the dominant market maker's pricing model on Polymarket's BTC 5-minute contracts: identified its price-feed composition via exclusive falsification testing, achieving ~6-tick out-of-sample replication of its quotes. Documented the methodology in a research paper.
- Derived a Brownian-probit fair-value model and a binary-CARA inventory-skew rule from first principles; validated against logged market data (within-slot $R^2 \approx 0.92$).
- Formalized the strategy as a Markov Decision Process to stress-test its structural limits, isolating queue-priority inaccessibility (not model quality) as the binding constraint on profitability — an honest negative result documented for rigor.
- Built and iterated weather-derivative strategies, developing new edges as market makers adapted — from a static bucket approach to a reactive trigger model to a probabilistic forecast engine blending a personal weather station, public forecasts, and live METAR data via Bayesian updating.
- Designed a World Cup cross-market latency-arbitrage strategy linking match markets to dependent group-advancement markets through Bayesian updating and Poisson goal modeling.
- Built low-latency multi-venue data infrastructure: direct WebSocket feeds (Binance, Coinbase, Kraken, Bitstamp), Chainlink oracle relay, and Polymarket CLOB, deployed on AWS (Ireland).

Volunteer Research Intern — Prof. Ali Bozbey's Group, TOBB ETÜ

Summer 2025

Superconducting / quantum-computing hardware · Ankara, Turkey

- Joined the group (which built Turkey's first superconducting quantum-computing hardware) on a volunteer basis after connecting with the team at ICSM; contributed to ongoing laboratory work over two weeks.

Engineering Intern — Ingenieurbüro Bickele & Bühler GmbH

Nov 2023

Electronics manufacturing & software · Stuttgart, Germany

- Worked across SMD pick-and-place, AOI, laser and soldering operations, incoming-goods QC, and device assembly; wrote software in C# and Arduino.

RESEARCH & WRITING

- Reverse-Engineering a Prediction Market Maker** — feed identification, volatility dynamics, and out-of-sample replication evidence from Polymarket BTC contracts. github.com/ErenEgeCelik
- Binary Market Making** — inventory-skew derivation under CARA utility, adverse-selection analysis, and empirical verdicts under realistic execution. github.com/ErenEgeCelik

ACTIVITIES

- Organizing Staff — International Conference on Superconductivity and Magnetism (ICSM) & ICSQMT, 2025 & 2026; supported conference operations alongside the organizing faculty.

SKILLS

Quantitative: probability & statistics, Bayesian inference, stochastic modeling, time-series analysis, market microstructure, derivatives pricing.

Technical: Python (pandas, NumPy), C#, low-latency WebSocket data systems, CLOB / REST APIs, Polygon / USDC / DeFi tooling, multi-agent coding workflows.